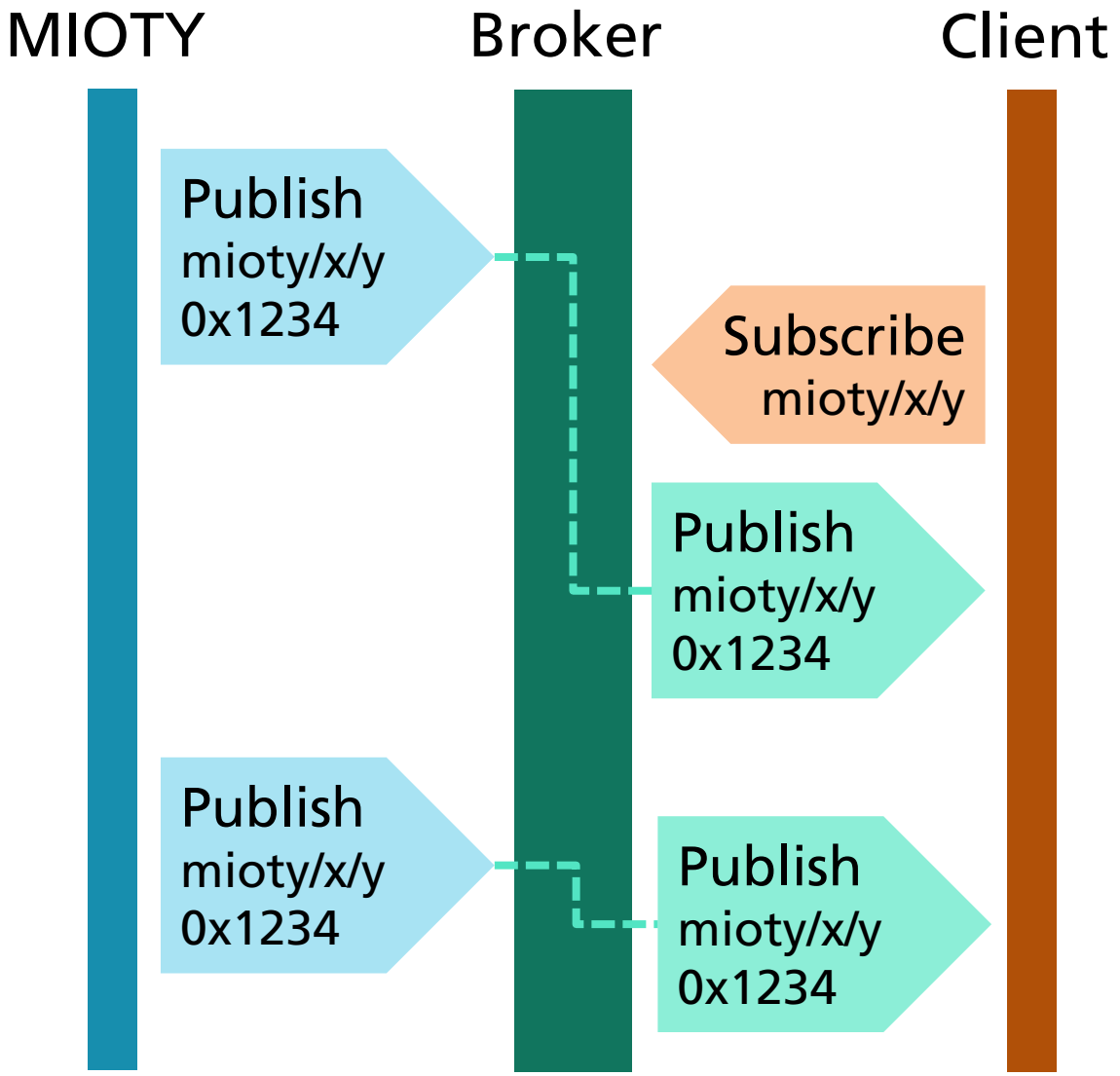
Mioty MQTT Interface

MQTT basics

- Data is organized in topics
- Topics can form a multi-level tree structure
- The MQTT broker distributes data between clients
- Clients can publish data to a topic
- Clients can subscribe to a topic to get published data



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MQTT End Point data topics

```
mioty/00-00-00-00-00-00-00-00/
                70-b3-d5-67-70-11-xx-xx/  uplink
                                          downlink
```

Common root for the Application Center

Or with wildcard: mioty/+/

End point specific subdirectory based on EUI

End point specific topics for uplink and downlink (BiDi only)

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MQTT uplink topic

`uplink` provides data and meta data as JSON object:

```
{
 "baseStations":   […],            Receiving base stations
 "cnt":            1234,           Packet counter
 "data":           [1,2,3,4],      Payload raw data
 "format":         0,              Payload format ID
 "typeEui":        1234567890,     Type EUI for blueprint selection
 "components":     {…},            Payload components (blueprint)
 "meta":           {…}             Meta information (blueprint)
}
```

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MQTT uplink topic base stations

`baseStations` contains one object for every receiving base station:

```
{
 "bsEui":     9876543210,
 "rssi":      -123.45,
 "snr":       6.78,
 "rxTime":    1234567890
}
```

Base station EUI	
RSSI in dBm	
SNR in dB	
Unix time of reception in ns	

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MQTT uplink topic components and meta

`components` is only present if a blueprint is provided for the end point and contains according interpreted payload components:

```
{
 "temperature": {"unit": "°C", "value": 22.5},
 "humindity": {"unit": "%rel", "value": 50},
 …
}
```

`meta` is only present if a blueprint is provided for the end point and contains according device meta information

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MQTT downlink topic

`downlink` accepts data and meta data as JSON object:

```
{
 "data":   [1,2,3,4],
 "format": 0,
 "queId":  0,
 "prio":   0.0,
 "presched": true,
 "cnt": [0,1,2,3]
}
```

`"data": [1,2,3,4],`	Payload raw data
`"format": 0,`	Payload format ID, optional
`"queId": 0,`	Queuing ID, optional
`"prio": 0.0,`	Priority, optional
`"presched": true,`	True to schedule immediately, optional
`"cnt": [0,1,2,3]`	Packet counter for prescheduling, optional

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MQTT End Point registration topics

`mioty/00-00-00-00-00-00-00-00/`

Common root for the Application Center

`register`

`deregister`

Operation specific topics

Mioty MQTT Interface

MQTT registration topic

`register` accepts end point information as JSON object:

```
{
 "eui":     12345678,
 "shAddr": 1234,
 "nwkKey": "0001…0e0f",
 "bidi":   true,
 "preAttach":true,
 "appKey": "0001…0e0f",
 "typeEui": 87654321
}
```

EUI	
Short address	
Network key	
True for bidi End Points, optional	
True to preattache End Point, optional	
Application key, optional	
Type EUI (blueprint), optional	

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MQTT deregistration topic

`deregister` accepts end point information as JSON object:

```
{
 "eui":    12345678
}
```

EUI

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MQTT subscription examples

- Uplink topic for sensor node 70-b3-d5-67-70-11-00-01:
  `mioty/+/70-b3-d5-67-70-11-00-01/uplink`
- All topics for sensor node 70-b3-d5-67-70-11-00-01:
  `mioty/+/70-b3-d5-67-70-11-00-01/#`
- Uplink topic for every sensor node:
  `mioty/+/+/uplink`
- Every topic for every sensor node:
  `mioty/#`
  or just everything:
  `#`

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